

Synergistic development in the Guangdong-Hong Kong-Macao Greater Bay Area: Index measurement and systematic evaluation based on industry-innovation-infrastructure-institution perspectives

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ABSTRACT

In recent years, promoting regional synergistic development has become the shared objective in city-regions during the new stages of development. This paper aims to uncover the dynamics of synergistic development in urban agglomerations by constructing an index measurement system based on four dimensions: industrial synergy, innovation synergy, infrastructure synergy, and institutional synergy. By comprehensively employing theories and methods such as “space of flow” big data analysis, network analysis, and interlocking network models, this study examines patterns and capacities of synergistic development in the Guangdong-Hong Kong-Macao Greater Bay Area. The main instances of industrial cooperation between cities are observed in Hong Kong-Shenzhen and Guangzhou-Shenzhen; capital flows primarily concentrate in Guangzhou-Shenzhen and Hong Kong-Shenzhen. Research paper and patent collaborations between cities are predominantly concentrated in Hong Kong-Shenzhen; joint laboratories are mainly established in Hong Kong-Macao and Hong Kong-Guangzhou; inter-city co-construction of higher education institutions is mainly focused on Guangzhou-Foshan and Shenzhen-Hong Kong. Infrastructure synergistic development performs best in Guangzhou-Shenzhen, Guangzhou-Foshan, and Shenzhen-Dongguan. Cross-border collaboration benefits from institutional advantages held by Hong Kong and Macao. Urban interactions with outstanding performance include Guangzhou-Shenzhen, Hong Kong-Shenzhen, Guangzhou-Foshan, Shenzhen-Dongguan. Additionally, increasing interaction trends can be observed between Guangzhou-Dongguan as well as Shenzhen-Zhuhai/Shenzhen-Foshan partnerships. Understanding the typical characteristics and potential trends of synergistic development within the GBA holds significant practical value for establishing a first-class bay area along with a world-class urban agglomeration.

1. Introduction

The city, as we know it but are still not familiar with it, is a complex system comprised of various subsystems such as society, economy, technology, and ecology (Acuto et al., 2018; Fang and Yu, 2017; Grimm et al., 2008; Li and Yi, 2020; Webb et al., 2018; Yin et al., 2023). Each subsystem is composed of capital, talent, technology, resources and other elements (Fig. 1). Local interactions, namely, the interactions of these elements within subsystem or among subsystems in one city is

intracoupling (Liu, 2017; Liu et al., 2018), which is called intra-city synergistic development at urban scale (Fig. 1). Under the action of regional interdependence (Liu and Zou, 2014; Moses, 1952; Rugman and Verbeke, 2004), the interactions of these elements and the sub-systems composed of these elements at the regional level between adjacent cities is pericoupling (Liu, 2017; Liu et al., 2018), which is called inter-city synergistic development at regional scale (Fig. 1). In addition, under the influence of globalization, informatization and digitization, the interactions at the regional level between distant cities

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With the continuous advancement of globalization and urbanization, synergistic development has become the primary choice for neighboring regions to enhance their core competitiveness and achieve mutual benefit and win-win results (Breul, 2020; Burger and Meijers, 2016; Chong and Pan, 2020; Taylor, 2005; Wang et al., 2022c). Taking the world-class bay areas (New York Bay Area, San Francisco Bay Area and Tokyo Bay Area) as examples, under the guidance of theoretical systems such as growth poles (Perroux, 1950), regional economic integration (Huh and Park, 2021), core-periphery (Krugman, 1991), regional interdependence (Liu and Zou, 2014), and global production networks (Yeung and Coe, 2015), a new path of regional synergistic development has been gradually formed through “sharing, collaboration, and complementarity”. The New York Bay Area is characterized by a spatial structure of “core areas leading - functional areas on the periphery”. In contrast, the San Francisco Bay Area has gradually strengthened its “pan-network” spatial structure through improvements in transportation infrastructure, while the Tokyo Bay Area is strengthening its “convective” spatial structure. The development and growth of

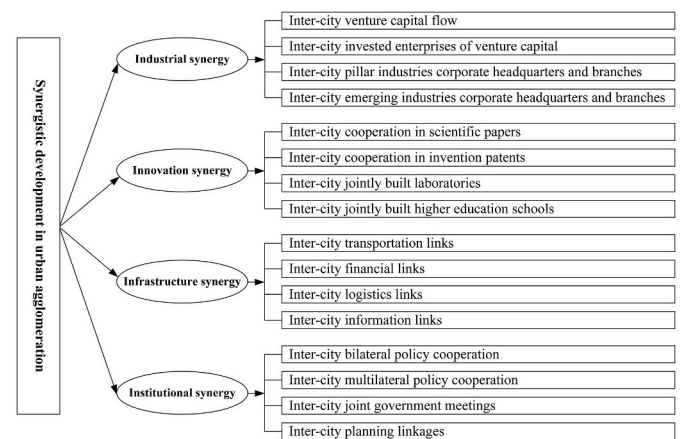


Fig. 2. Index system for the synergistic development.

With regard to the understanding of regional synergistic development, [Fang \(2017\)](#) takes Beijing-Tianjin-Hebei as an example and illustrates that the connotation of synergistic development of urban agglomerations is to promote the coordination of planning, transportation, industry, market, technology, finance, information, ecology and environment. [Zhou et al. \(2018\)](#) takes synergistics as a reference and argues that regional synergistic development means that adjacent cities in a region break through the constraints of administrative divisions through the interactions of technology, investment and information etc., optimize the allocation of factors and resources, and then maximize the overall regional benefits. [Shi et al. \(2019\)](#) proposes the development of “geographical synergy”, namely, the mechanism, process and dynamics of the Earth’s surface system and regional sustainability, aiming to realize the transformation of geographical research from the explanation of human-nature interactions to the design of human-nature synergy. [Yin et al. \(2023\)](#) proposes a novel network-based approach that combines qualitative analysis and quantitative verification to assess the synergistic development of urban agglomerations in the Yangtze River Delta Urban Agglomeration. The synergistic development of urban agglomerations has shifted from the previous study on capacities based on “space of place” to the study on relationships based on “space of flow”. We believe that the synergistic development of urban agglomerations means the efficient interaction and optimal allocation of capital,

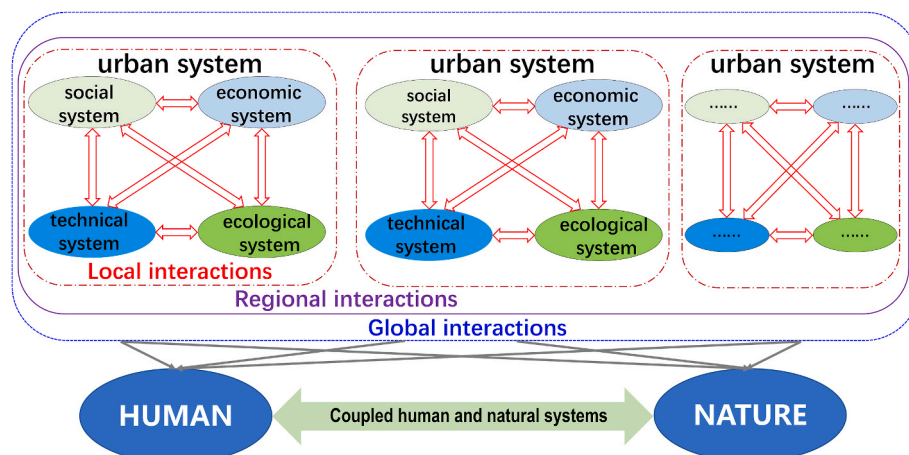


Fig. 1. Framework of synergistic development in global city region.

resources, talents, technology and other elements between/among and within cities, especially the cooperation and collaboration between cities.

In recent years, with the promulgation and implementation of a series of national strategies, such as the Outline of the Beijing-Tianjin-Hebei Synergistic Development Plan, the Outline of the Guangdong-Hong Kong-Macao Greater Bay Area Development Plan, and the Outline of the Yangtze River Delta Regional Integration Development Plan, promoting regional synergistic development has become the common purpose of the new development stages (Fang, 2017; Zhou et al., 2018). However, as the three major urban agglomeration regions with the highest degree of openness, the largest economic vitality and the strongest innovation ability in China, there are still problems such as administrative segmentation, industrial homogeneity, competition homogeneity, and resource mismatch etc. In other words, the mechanism of cross-regional cooperation, sharing, protection and governance is not yet sound, and synergies and connectivities need to be improved. From the Pearl River Delta urban agglomerations to the Guangdong-Hong Kong-Macao Greater Bay Area (GBA), from the “front shop and back factory” model since the reform and opening up to the current “the largest metropolitan belt in East Asia” and then to the “world-class Bay Area and world-class urban agglomerations” under construction, the GBA has become a successful practice of realizing the “multi-win” situation of regional economy through industrial cooperation (Wang et al., 2022a; Xu et al., 2023; Ye et al., 2019; Zheng et al., 2022). However, in the new period, new stage and new situation, as a special area of “one country, two systems and three tariffs”, the GBA is a typical case of studying regional synergistic development. Only by objectively and rationally analyzing the particularity and typicality of the GBA can we accurately grasp the possible and impossible aspects of its synergistic development. The typicality of the GBA is mainly manifested in the following aspects: particularity of institutional environment (socialism, capitalism); complexity of relationship scale (central-local relationship, intergovernmental relationship); significance of internal differences (economic differences, cultural differences, etc.). As a result, the development of the GBA faces many challenges, such as the mismatch between supply and demand structure, tightening constraints on resources and energy, increasing pressure on the ecological environment, and unequal public services etc. There have been studies on the synergistic development of the GBA, mainly reflected in the strategic discussion of the synergistic mechanism (Liu et al., 2022; Zhou et al., 2018), regional integration (Liu et al., 2020b; Wang et al., 2023), industrial synergy model (Chong and Pan, 2020; Liu, Yi et al., 2020a; Wang et al., 2022a; Xu et al., 2022), innovation linkages (Ma et al., 2021, 2022), cross-border commuting (Zhang et al., 2020), environmental synergistic governance (Wang et al., 2022b; Xu and Ma, 2020). Synergistic development research has shifted from “space of place” based on capabilities to “space of flow” based on relationships, focusing more on connectivity and mobility between cities. Spatial network has become an important perspective to understand the internal relations of urban agglomerations. Based on the improved gravity model and social network analysis method, Peng (2017) revealed that the network connections in the GBA performed a gradient decay trend from the core of Hong Kong-Shenzhen-Guangzhou to the peripheral cities, accompanied by significant characteristics of core peripheral structure. By adopting “Baidu Index”, Chen (2018) identified the connectivities among 11 cities in the GBA, and explained that “Hong Kong-Shenzhen” and “Guangzhou-Foshan” are the two poles of factor agglomeration. Based on the analysis of traffic flow and information flow, Qiu et al. (2019) revealed the spatial network patterns of the GBA with Guangzhou-Shenzhen as the dual core from the perspective of “space of flow”. Existing studies have paid too much attention to the imagination of the “three cities” in Hong Kong, Guangzhou and Shenzhen, and paid insufficient attention to how to create a good situation of efficient and convenient flow of factors, further form collaborative and inclusive intra-regional urban networks. After all, the synergistic development of urban agglomerations is

ultimately not to eliminate intra-regional differences or achieve intra-regional averages, but to enhance the overall connectivity, synergy and competitiveness in city regions. The dynamics of the synergistic development of the Guangdong-Hong Kong-Macao Greater Bay Area in the new era is more intricate, encompassing a wider range of fields, featuring diverse models, and posing greater challenges in terms of quantification. However, there is still a lack of systematic quantitative analysis on the overall evaluation and key constraints of the synergistic development of the GBA. Currently, the evaluation of synergistic development in urban agglomerations primarily focuses on assessing the synergistic capabilities among different cities within these agglomerations. However, there is insufficient attention given to evaluating the synergistic relationships between these cities. Additionally, some indicators lack certain mathematical properties in their design. Therefore, it becomes necessary to conduct a comprehensive assessment of the level of synergistic development in the Greater Bay Area to promote its overall synergy. At the same time, the construction of more targeted synergistic measurement indicators is the key to achieving comprehensive evaluation. Zhou et al. (2018) illustrates that the synergistic development of urban agglomerations is mainly affected by four major forces, such as resource endowment differences, capital market, government institutional arrangements, and external environment changes. Ye et al. (2022) examines that the synergistic development of the GBA is undergoing mechanism changes at four levels, which are mainly reflected in the global shift, the internal and external market integration, the governance rescaling, and the construction of new infrastructure. This study intends to construct an evaluation index system for synergistic development from four dimensions: industrial synergy, innovation synergy, infrastructure synergy and institutional synergy. Furthermore, this research comprehensively adopts theories and methods such as “space of flow”, big data, network analysis and interlocking network model to conduct a quantitative analysis of the relationships, mobility and connectivity among cities in the GBA. However, our research incorporates prior studies and makes the following contributions: Firstly, we examine the concept and connotation of synergistic development in urban agglomeration areas based on theories such as “space of flow” and network analysis. Secondly, we employ big data and interlocking network model to establish indicators and methods for evaluating synergistic development in urban agglomeration areas. Thirdly, we provide explanations regarding patterns and capacities of synergistic development at the urban scale with the aim of further guiding regional development.

2. Methods

2.1. Index system construction of synergistic development of urban agglomerations

The key to the establishment of indicator system for the synergistic development is to solve the problem of “what to measure” and “how to measure” (Fan et al., 2019; Li and Yi, 2020; Yuan et al., 2017; Zhang et al., 2022). The United Nations Sustainable Development Goals (SDGs) consist of 17 sustainable development goals and over 200 specific indicators, encompassing multiple dimensions including society, economy, environment, and policy (Masuda et al., 2022; Yamasaki and Yamada, 2022). They serve as a crucial reference for this paper in establishing a synergistic development indicator system. The SDGs adopt the principles of classification, grading, and stratification to observe the developmental trends of various countries/regions through diachronic data estimation. Their aim is to comprehensively address the development issues pertaining to social, economic, and environmental dimensions in an integrated manner. The SDGs emphasize the measurement of scale, level, and capacity while also focusing on scientific and technological cooperation as well as South-South cooperation and partnerships among countries. The Global Innovation Index (GII), published by the World Intellectual Property Organization (WIPO), Cornell

University, and INSEAD, and an overall GII score is calculated as a simple average of input and output sub-index scores. Input indicators include: (1) institutions; (2) human capital and research; (3) infrastructure; (4) market maturity; and (5) business maturity. Output indicators comprise: (6) knowledge and technology output; along with (7) creative output. The GII primarily measures the agglomeration capacity of innovation factors with technical cooperation and innovation networks being integral components within the innovation ecosystem. Led by East China Normal University, the *report on index of urban coordinated development capability in the Yangtze River Economic Belt* has been released. Drawing on scientific theories such as growth poles, regional coordinated development, and regional resilience, an evaluation index system comprising 4 factor levels and 18 specific indicators has been established. The ability of urban synergistic development is comprehensively examined from four perspectives: economic development, scientific and technological innovation, exchange services, and ecological support. The China Development Research Foundation has unveiled the integrated development index of urban agglomerations, constructing an index system to characterize regional integration from four aspects: economic agglomeration degree (A), regional connectivity (C), economic equalization (E), and policy synergy (P). This forms the ACEP index. The measurement of regional integration relationship focuses on assessing the connection between cities. The Globalization and World Cities Research Network (GaWC) examines the global distribution of advanced producer services to assess cities' connectivity on a global scale. It measures a city's Global Network Connectivities based on its extensive presence of productive services such as finance, advertising, law, accounting, and management consulting. GaWC analyzes the worldwide allocation of advanced producer services to position cities in terms of their connectivity within the global network. The network model developed by the GaWC (Derudder et al., 2010; Taylor, 2005; Taylor et al., 2002), which is utilized to assess global connectivities, serves as a crucial framework for constructing relational indicators in this study. However, GaWC primarily focuses on urban connectivities based on advanced producer service (APS) firms (Derudder et al., 2010), making it challenging to adequately capture the current manufacturing advantages of the GBA. Therefore, based on a comprehensive understanding of the Sustainable Development Goals (SDGs) and Global Innovation Index (GII), we construct a comprehensive evaluation index system for synergistic development in the Greater Bay Area. This system incorporates the calculation method of GaWC and aligns with the GBA's development trend. It focuses on measuring the one-to-one, one-to-many, and many-to-many relationships among cities in terms of industrial synergy, innovation synergy, infrastructure synergy, and institutional synergy.

Industrial synergy provides economic support for the development of the Guangdong-Hong Kong-Macao Greater Bay Area's global layout. Innovation synergy is an inherent requirement for establishing the Greater Bay Area as an international technological innovation center. Infrastructure synergy embodies the service concept in building a high-quality living circle within the Greater Bay Area. Institutional synergy serves as a policy guarantee to establish a world-class Bay Area.

The indicators of industrial synergy encompasses the magnitude of venture capital flow between cities, the number of venture capital firms operating across cities, the level of connectivity between headquarters and branches in pillar industries, as well as the degree of interconnection between headquarters and branches in emerging industries within urban clusters. For instance, it quantifies the total investment amount and number of investment enterprises engaged in start-up venture capital activities in City B by City A within an urban agglomeration during a given year. The time unit is annual, based on data sourced from China's Investment and Financing Events Database (see: <https://data.cvsources.com.cn/cn/login>). Utilizing an interlocking network model, we calculate the strength of connection between City A - where electronic information industry and biomedical enterprises are headquartered - and City B - where their respective branches are located within an urban

agglomeration. Time unit is measured annually, primarily relying on data derived from either Industrial and Commercial Enterprise database or Enterprise Registration Database specific to the region encompassing the urban agglomeration.

The innovation synergy encompasses the quantity of scientific and technological paper collaborations among cities, the number of patent collaborations between cities, the count of joint laboratories established by cities, and the number of colleges and universities jointly built between cities. Scientific and technological papers authored within each city in the urban agglomeration are collected, and collaborative papers between cities are categorized based on their respective addresses. The primary data sources include Web of Science database and China CNKI database. Invention patents filed with each city as the author's address within the urban agglomeration are compiled, and cooperative patents between cities are classified according to their addresses. Data for this analysis originate from statistical records provided by China National Intellectual Property Administration. The number of jointly operated schools among various cities within an urban agglomeration refers to educational institutions managed by City A located in City B during a specified year; official education department statistics serve as data sources for this information collection process. Similarly, co-established laboratories among different cities within a city cluster refer to laboratories jointly set up by City A in City B during a specified year; laboratory data published by official science and technology departments along with Guangdong-Hong Kong-Macao Greater Bay Area Joint Laboratory records contribute to these findings.

The concept of innovation synergy encompasses the intensity of traffic flow, financial connection, logistics connection, and information flow between cities. This includes both the traffic flow within an urban agglomeration as well as between specific cities within that agglomeration during a specified time period. The daily driving migration index released by Amap serves as an indicator of commuting intensity among the population in the Guangdong-Hong Kong-Macao Greater Bay Area, with higher values indicating a greater number of cross-city driving users. To assess the financial connection strength between city A and city B, network data from banks in each city within the urban agglomeration is selected and analyzed using an interlocking network model based on their service value (i.e., total financial links). The maximum normalized processing result represents the level of financial synergy between these cities. The bank POI data includes 24 bank companies such as Bank of China, Industrial and Commercial Bank of China, China Construction Bank, Agricultural Bank of China, Bank of Communications, China Merchants Bank, Postal Savings Bank of China, Industrial Bank, China Minsheng Bank, China Citic Bank, Everbright Bank, Ping An Bank, Huaxia Bank, China Guangfa Bank, Bank of Guangzhou, HSBC Bank, Hang Seng Bank, Standard Chartered Bank (Hong Kong), Bank of East Asia, Citibank, Chong Hing Bank (Hong Kong), BPOP (Taiwan), Luso International Banking, Banco Nacional Ultramarino, etc. Similarly, for assessing logistics connection strength, network data from logistics companies in each city is used to calculate the service value-based (i.e., total logistics links) logistics synergistic relationship between city A and city B using an interlocking network model. The logistics POI data includes seven express companies such as SF Express, China Post, Deppon Express, STO Express, YTO Express, ZTO Express and YUNDA Express, including the address of the logistics site, the name of the express company. Finally, a connection matrix is constructed based on Baidu Index to directly reflect the information connection intensity between cities; here again, the maximum normalized processing result of search volume represents their information synergistic relationship.

Institutional synergy encompasses the extent of bilateral policy cooperation, multilateral policy collaboration, joint meetings, and planning connections between cities. The intensity of policy correlation is characterized by the number of policy documents, which are statistically analyzed to determine the level of bilateral and multilateral policy correlation among cities. A statistical analysis is conducted on government planning documents issued by cities in the GBA, covering

areas such as industrial parks, talent flow, infrastructure development, environmental co-governance, etc., to calculate the number of documents related to planning connectivity between cities. The frequency of key conferences like the Guangdong-Hong Kong Joint Conference, Guangdong-Macao Joint Conference and Pan-Pearl River Delta Regional Co-operation Chief Executives' Joint Conference is analyzed with a focus on topics including industry, economy and trade, infrastructure development, environmental protection and social services. Based on the topological relationship between various policy subjects respectively established correlation matrices for corresponding cities are used to represent the intensity of policy correlation through counting the number of relevant policy documents. These policy documents mainly originate from sources such as the Policy Database provided by Guangdong Provincial Leading Group Office for Promoting Development in Guangdong-Hong Kong-Macao Greater Bay Area website and other local government websites.

2.2. Comprehensive measurements of synergistic development of urban agglomerations

2.2.1. Statistical model of factor flow

The former urban spatial network focuses on the urban connection based on “space of place”, while the “space of flow” generated by the flow of elements emphasizes the relationship and network. The “space of flow”, first proposed by Castells (2004), has gradually become an important perspective for understanding and reconstructing urban network structure (Castells, 2009). “Space of flow” is a unidirectional, bidirectional or multi-directional flow network dynamic space composed of population flow, traffic flow, information flow, capital flow, technology flow, knowledge flow and other elements. At present, the “space of flow” research based on big data and information technology not only includes the flow of factors between adjacent cities, but also includes the interconnections between distant cities worldwide.

The statistics of factor flow are mainly applied to the Origin-Destination (O-D) data and non Origin-Destination (non-O-D) data in the evaluation index system. For O-D data, the $n \times n$ asymmetric O-D flow matrix of n cities in urban agglomeration is mainly constructed. For non-O-D data, $n \times n$ symmetric O-D flow matrix of n cities in urban agglomeration is mainly constructed.

The statistical method of factor flow for O-D data is shown in Equation (1):

$$C_{ab} = C_{a \text{ to } b} + C_{b \text{ to } a} \quad (1)$$

Where C_{ab} represents the factor flow between city a and city b . Taking investment flow, traffic flow and other typical O-D data as an example, $C_{a \text{ to } b}$ represents the total amount of investment from City a to City b ; $C_{b \text{ to } a}$ represents the total investment of City b into City a .

The statistical method of factor flow for non-O-D data is shown in Equation (2):

$$C_{ab} = C_{a \text{ and } b} \quad (2)$$

Where C_{ab} represents the factor flow between city a and city b . Taking typical non-O-D data such as paper cooperation as an example, $C_{a \text{ and } b}$ represents the amount of paper cooperation between city a and city b .

2.2.2. Urban network analysis

Urban network analysis mainly focuses on quantifying the relationship between large, medium and small cities at different spatial scales, which can be divided into directed network and undirected network, as well as relation-based and location-based perspectives (Wang et al., 2022c, 2023). The difficulty lies in the collection and acquisition of urban relationship data and the formation of urban relationship matrix to reflect the real relationship between cities (Fang et al., 2020; Liu and Derudder, 2013; Liu, Yaolin et al., 2020). Based on the principle of dominant flow analysis, this paper explores the gross links of each urban

node in the network, including flow analysis, flow direction analysis and flow strength analysis, etc., and then reveals the flow characteristics of population flow, traffic flow, capital flow, technology flow and information flow among different urban nodes.

2.2.3. Interlocking network model

The interlocking network model was proposed by the GaWC led by Taylor (Taylor, 2001, 2005; Taylor et al., 2002). This model is based on the distribution of corporate headquarters and branches in APS. In the model, it is assumed that there are m APS enterprises in n cities, and the value of an enterprise in a city is measured by the service value of the enterprise in the city, and the importance of the enterprise is mainly scored by the service value of the enterprise in the city, which is represented by V_{ij} (Taylor et al., 2002). V_{aj} , $a \leq n, j \leq m$ represents the service value of enterprise j in city a , V_{ij} constitutes the $n \times m$ service value matrix V (Taylor et al., 2002). The basic nodal connection (R_{abj}) between city a and city b through enterprise j can be obtained from the service value matrix V , namely, $R_{abj} = V_{aj} \times V_{bj}$. R_{abj} is the elemental interlock between city a and City b . Therefore, $R_{ab} = R_{ab1} + R_{ab2} + R_{abj} + \dots + R_{abm}$ ($1 \leq j \leq m$). R_{ab} represents the sum of elemental interlock between city a and city b through all enterprise units, that is, city interlock, which represents connectivity between cities.

3. Systematic evaluation of synergistic development in the Guangdong-Hong Kong-Macao Greater Bay Area

3.1. Industrial synergy

Venture capital, is mainly a financing method to provide financial support for start-ups, which more represents the future direction of industrial development and the trend of technological change. Based on the database of China's inter-city venture capital flow, the information of the invested enterprises, investment institutions, investment amount, investment year, investment rounds and other information in the GBA was sorted out. Total investment was the sum of the real investment amount and investment rounds, and more than 2000 effective investment relationships were obtained to further aggregate the investment data at the city scale. Total investment of venture capital in the GBA was mainly performed in the investment of cities into cities themselves. Total investment was mainly concentrated in Shenzhen, whose total investment of venture capital was close to 50% of the total amount of the GBA. The inter-city venture capital was mainly concentrated in Guangzhou-Shenzhen and Hong Kong-Shenzhen, and the investment from Shenzhen to Guangzhou and the investment from Hong Kong to Shenzhen were mainly. This investment pattern was followed by investment from Hong Kong to Foshan, Guangzhou to Zhuhai, Shenzhen to Dongguan, and Shenzhen to Zhongshan (Fig. 3 a). Venture capital events in the GBA were mainly concentrated between Guangzhou and Shenzhen. Although the total amount of venture capital investment in Guangzhou-Shenzhen and Hong Kong-Shenzhen was roughly the same, there were more than 300 venture capital enterprises between Guangzhou and Shenzhen, and less than 50 venture capital enterprises in Hong Kong and Shenzhen, which further indicated that small and medium-sized enterprises were important carriers for the implementation of technological innovation in the GBA (Fig. 3 b). Since the Outline of the Plan for Reform and Development of the Pearl River Delta (2008–2022) compiled in 2008, the three metropolitan areas of “Guangzhou-Foshan-Zhaoqing”, “Shenzhen-Dongguan-Huizhou” and “Zhuhai-Zhongshan-Jiangmen” have become the distinctive features of the urban network structure of the Pearl River Delta in a long time. However, the administrative boundary between cities has gradually blurred, after the integration of Guangzhou and Foshan, Shenzhen and Foshan linkage trend appeared. Recent data show that Shenzhen's leading enterprises such as *China Resources* (华润), *FRD* (飞荣达) and *CIMC Tianda* (中集天达) were laying out in Foshan, and the GBA was building a new type of urban relationship.

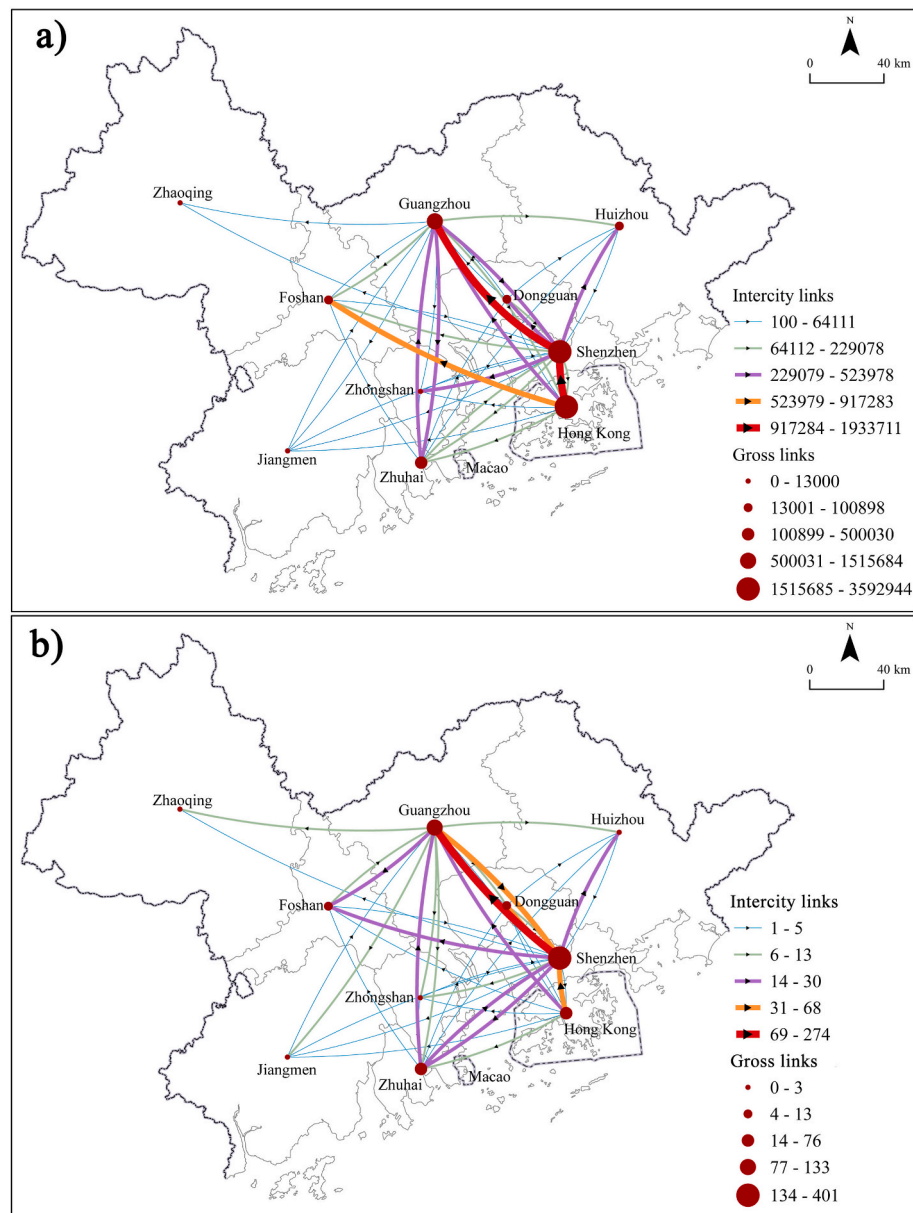


Fig. 3. Total investment of venture capital (a) and total invested enterprises of venture capital (b) in the GBA.

With the new generation of electronic information industry as the representative industry of the pillar industries in the GBA, the inter-city corporate headquarters branches were mainly reflected in Hong Kong-Shenzhen and Shenzhen-Dongguan, followed by Hong Kong-Dongguan, Shenzhen-Huizhou (Fig. 4 a). As the hub of the electronic information industry in the GBA, Shenzhen has gathered the headquarters of a number of leading domestic enterprises such as Huawei and ZTE. It is not only a radiation source to drive the development of the electronic information industry in other cities, but also a magnetic field to attract investment in the electronic information industry in other cities in the GBA. Under the “front shop and back factory” model, Hong Kong mainly played the role of strategic decision-making, international contacts and market expansion. Strategic pillar industries have basically formed the Guangzhou-Shenzhen-Hong Kong Science and Technology Innovation Corridor as the main axis. With biotechnology industry as the representative industry of emerging industries in the GBA, the inter-city corporate headquarters branches were mainly reflected in Shenzhen-Guangzhou and Hong Kong-Shenzhen, Shenzhen-Dongguan, followed by Shenzhen-Zhuhai, Hong Kong-Guangzhou. Shenzhen, Guangzhou

and Hong Kong have become biotechnology industry hubs in the GBA with their outstanding R&D capabilities and excellent innovation environments. Zhuhai and Dongguan, as sub-centers, mainly cultivated enterprises in specific specialized fields and played the function of production bases. Strategic emerging industrial linkages are mainly concentrated in the three central cities of Guangzhou, Shenzhen and Hong Kong in the GBA.

3.2. Innovation synergy

Based on the international journal database (Web of Science) and domestic journal database (CNKI), the urban-scale paper cooperation relationships in the GBA were revealed (Fig. 5 a). The evolution trends of paper cooperation in the GBA were as follows: Since 2000, papers have been co-published mainly in Hong Kong and Guangzhou; After 2005, the amount of co-published papers between Shenzhen and other cities increased rapidly, mainly due to the fact that in order to make up for the shortage of local higher education resources, domestic and foreign famous universities were attracted to run schools in Shenzhen; Since

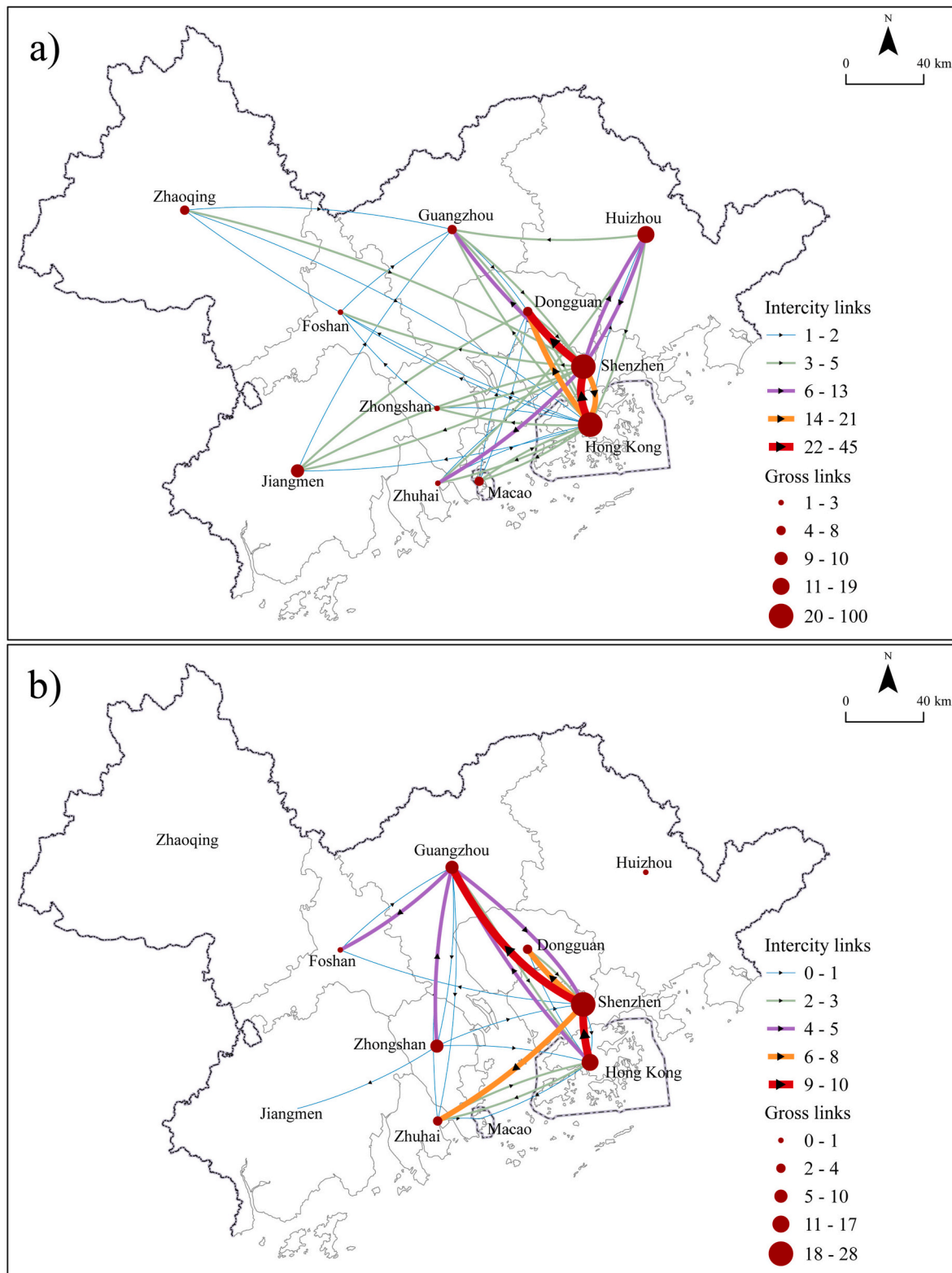


Fig. 4. Inter-city corporate headquarters branches of electronic information industry (a) and biotechnology industry (b) in the GBA.

2010, Guangzhou, Hong Kong and Shenzhen have taken the lead in the co-publication of papers. After 2014, the amount of co-published papers between Hong Kong and Shenzhen exceeded that between Hong Kong and Guangzhou, and the amount of co-published papers between Guangzhou and Shenzhen also showed a growing trend. At present, paper cooperation in GBA was mainly reflected in Hong Kong and

Shenzhen, followed by Hong Kong-Guangzhou and Shenzhen-Guangzhou, and then Guangzhou-Foshan, Guangzhou-Dongguan, Guangzhou-Zhuhai and Guangzhou-Huizhou. Under the influence of factors such as geographical proximity and institutional proximity, the cooperation among researchers and research institutions have helped the GBA to form a “Guangzhou-Shenzhen-Hong Kong” multi-center

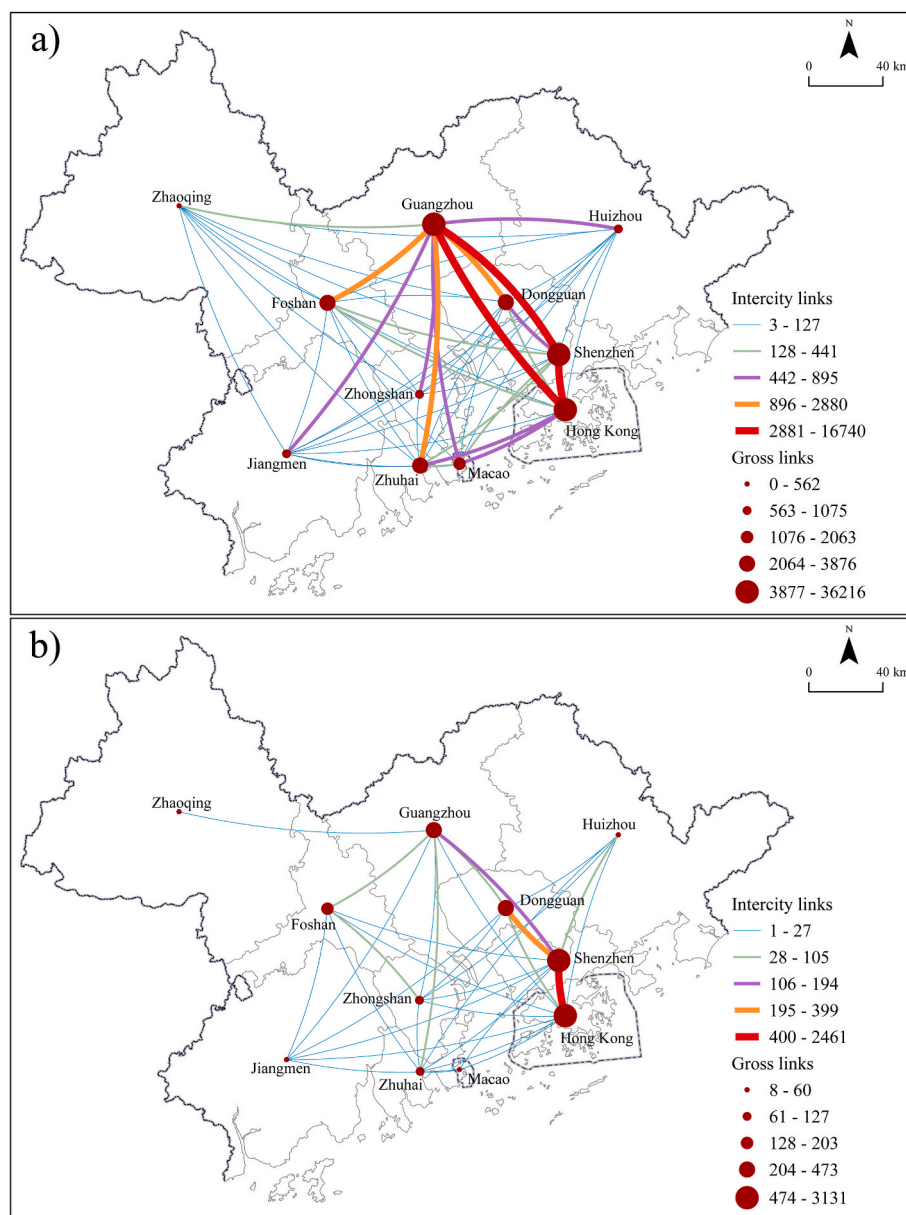


Fig. 5. Inter-city scientific paper cooperation (a) and patent cooperation (b) in the GBA.

synergistic innovation community. Total amount of patent applications in the GBA has obvious advantages on global scale. According to the Global Innovation Index, the amount of PCT applications filed by Shenzhen-Hong Kong-Guangzhou ranks second in the world after Tokyo-Yokohama. Patents in the GBA were mainly concentrated in the electronic information industry, followed by green petrochemical industry and advanced materials industry. Across the country, Beijing had the most patent cooperation with the Shenzhen-Hong Kong-Guangzhou cluster, and the most cooperative applicant was Huawei. Within the GBA, patent cooperation applications between cities were mainly concentrated in Hong Kong and Shenzhen, followed by Shenzhen and Dongguan, Shenzhen and Guangzhou, and Foshan and Zhongshan (Fig. 5 b).

The Department of Science and Technology of Guangdong Province started the construction of Guangdong-Hong Kong-Macao Joint laboratories in 2019. At present, twenty joint laboratories that have been built were led by Guangdong and jointly built by universities, scientific research institutions and other legal entities of Hong Kong and/or Macao with basis for cooperation. Thanks to the concerted efforts of the

Guangdong, Hong Kong and Macao teams, Guangdong-Hong Kong-Macao Joint laboratories have made good progress in the areas of scientific research achievements, transformation and application, introduction of talent teams, and scientific and technological cooperation and exchanges. Guangdong-Hong Kong-Macao Joint laboratories were mainly reflected in Hong Kong-Macao and Hong Kong-Guangzhou. Hong Kong and Macao participated in the co-construction of eleven joint laboratories. Hong Kong and Guangzhou participated in the co-construction of ten joint laboratories. Hong Kong and Shenzhen, Guangzhou and Macao, Shenzhen and Macao jointly build seven, six and five laboratories respectively. There were only two joint laboratories in Guangzhou and Shenzhen (Fig. 6 a). Cooperation in running schools is an important measure to broaden enrollment channels and promote education integration. Inter-city co-construction of higher education schools was mainly concentrated in Guangzhou and Foshan, Shenzhen and Hong Kong. Inter-city co-construction of higher education schools between Guangzhou and Foshan were mainly Sun Yat-sen University, South China University of Technology, South China Normal University and Southern Medical University, which were located in Guangzhou, to

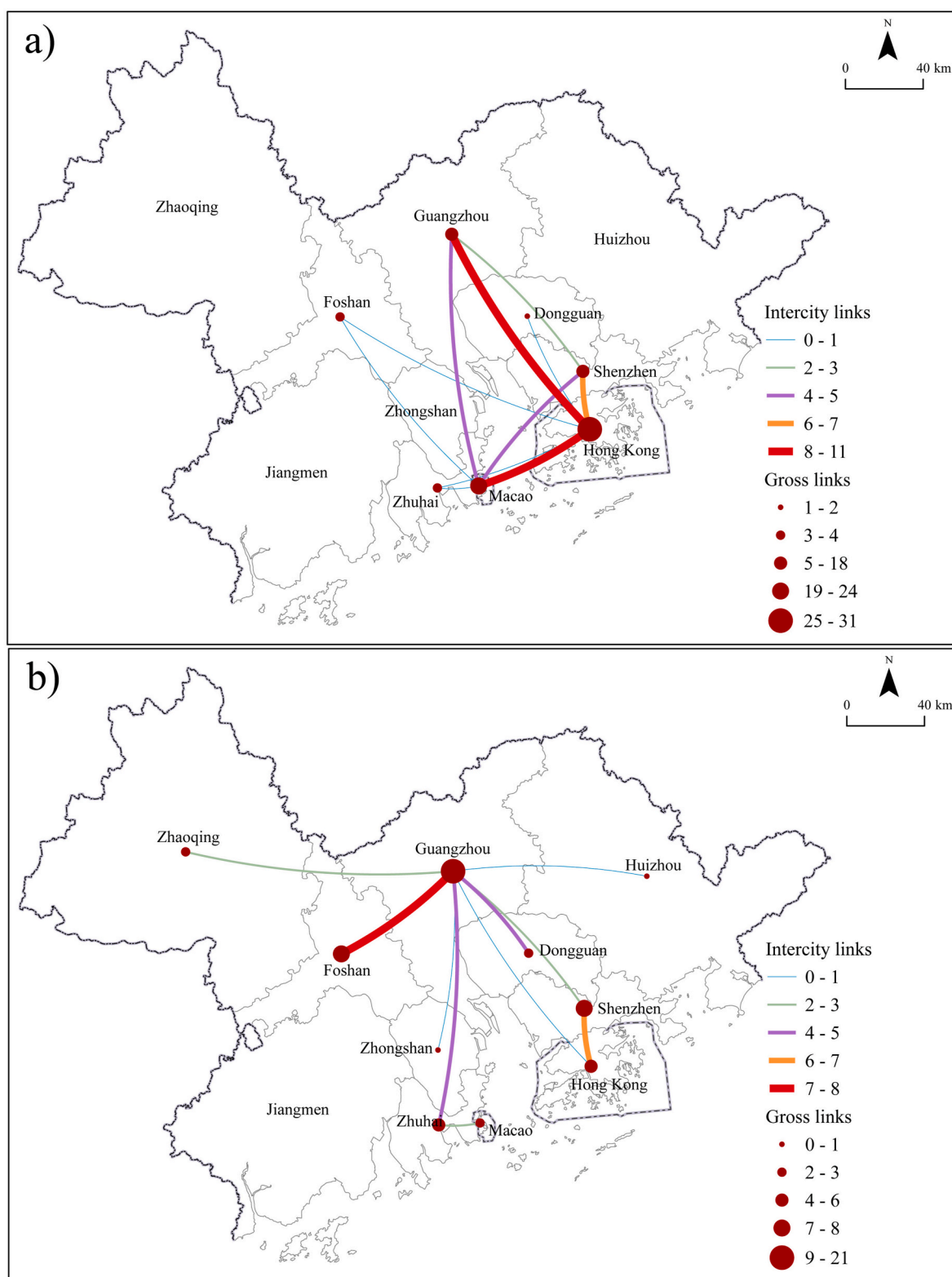


Fig. 6. Guangdong-Hong Kong-Macao Joint laboratories (a) and co-construction of higher education schools (b) in the GBA.

set up new campuses in Foshan. Meanwhile, the university of Hong Kong, the Hong Kong Polytechnic University, the Chinese University of Hong Kong, the Hong Kong Baptist University and the Hong Kong University of Science and Technology have established institutes in Shenzhen (Fig. 6 b).

3.3. Infrastructure synergy

Traffic flow in the GBA was reflected by the daily driving migration index released by the Amap. Shenzhen-Dongguan route and Guangzhou-Foshan route had the highest single-day migration index, and the actual migration scale between Shenzhen-Dongguan route and Guangzhou-

Foshan route was roughly the same, followed by Guangzhou-Dongguan route, Shenzhen-Huizhou route and Dongguan-Huizhou route, and the actual migration index from Dongguan to Guangzhou and Huizhou to Shenzhen was stronger than the actual migration index from Guangzhou to Dongguan and Shenzhen to Huizhou (Fig. 7 a). From the analysis of the actual migration index, the population flow between Shenzhen and Dongguan was more frequent. Benefiting from strong transportation infrastructure service capabilities, Shenzhen and Dongguan were connected by nine expressways, and Guangzhou and Foshan were mainly connected by six expressways. The strongest logistics links in the GBA were performed between Guangzhou and Shenzhen, which were much stronger than other cities, followed by Guangzhou-Foshan, Shenzhen-Dongguan, Guangzhou-Dongguan, Hong Kong-Guangzhou, Shenzhen-Foshan, and Hong Kong-Shenzhen (Fig. 7 b).

The inter-city information flow based on Baidu index was mainly concentrated in Guangzhou-Shenzhen, Shenzhen-Dongguan, Guangzhou-Foshan, followed by Hong Kong-Guangzhou, Hong Kong-Shenzhen, Shenzhen-Foshan, Guangzhou-Dongguan (Fig. 8 a). With the support of the interlocking network model, the total financial links

between cities were calculated (i.e. the connectivity between cities) (Wang et al., 2023). The highest degree of financial links among cities was between Guangzhou and Shenzhen, Guangzhou and Foshan, followed by Guangzhou and Dongguan, Shenzhen and Foshan. Cities with the largest amount of bank branches in the GBA were Guangzhou, Shenzhen, Foshan, Hong Kong and Dongguan, in order. Hong Kong, as the international financial center, has the most bank outlets in the order of Bank of China, HSBC, Hang Seng Bank, Bank of East Asia and Standard Chartered Bank. Except Bank of China, other banks have relatively few branches in the nine cities of the Pearl River Delta, resulting in the closeness of internal financial links far lower than the external financial links.

3.4. Institutional synergy

Under “one country, two systems”, the decisive influence of institutional factors on the synergistic development of the GBA was obvious. The policy documents published on urban official website of the GBA (involving joint meetings, economic and trade agreements, inter-city

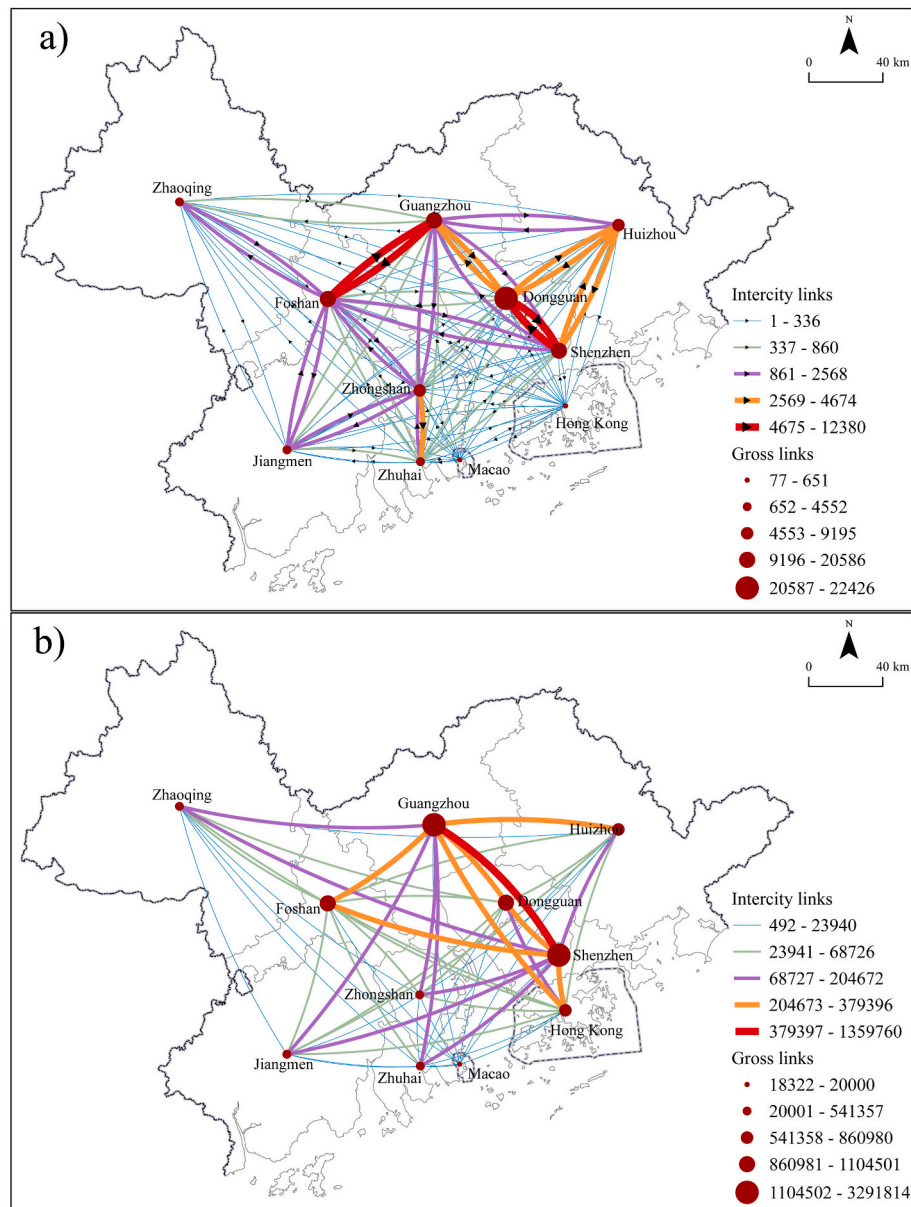


Fig. 7. Traffic flow (a) and logistics based on POI data (b) in the GBA.

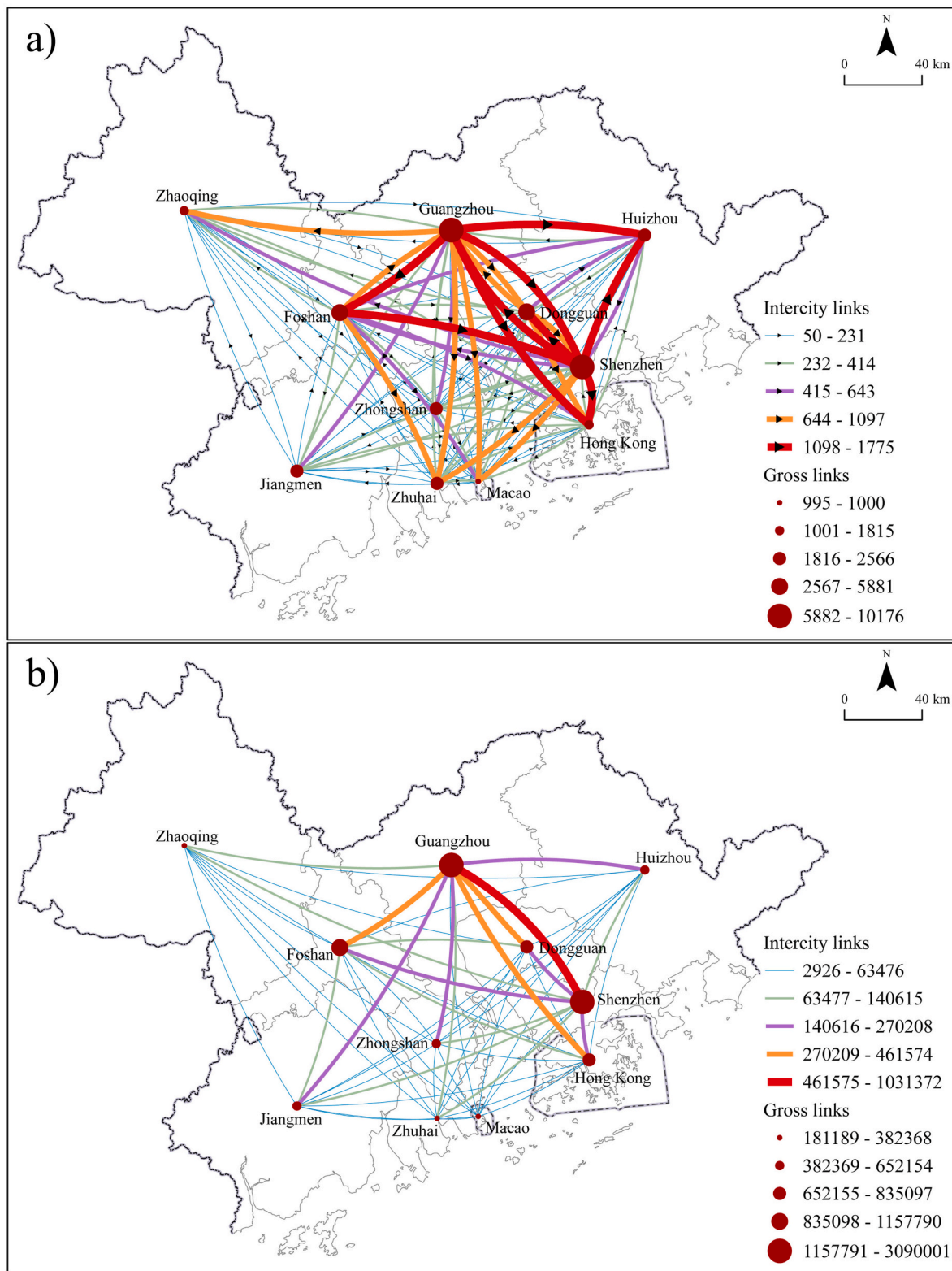


Fig. 8. Information service (a) and financial service based on bank POI data (b) in the GBA.

planning convergence, cross-border regional cooperation, inter-city transportation infrastructure, etc.) were used to build database to form quantitative indicators of policy interactions among urban government entities, focusing mainly on the measurement of bilateral and multilateral relations between urban entities (Li et al., 2022). For example, the Guangdong-Hong Kong Joint Meeting, the Guangdong-Macao Joint

Meeting and the Pan-Pearl River Delta Regional Co-operation Joint Executive Chiefs Meeting, which have been running for many years, mainly focus on industry, economy, trade, infrastructure, environmental protection and social services etc. Four central cities of Hong Kong, Macao, Guangzhou and Shenzhen had different policy links with other cities in the GBA.

Neighboring cities paid more attention to bilateral policy cooperation, and mainly focused on industrial synergy and infrastructure synergy policies, including cross-border synergistic development and transportation infrastructure connections etc. Bilateral policy cooperation between cities was mainly concentrated in Hong Kong and Shenzhen, Macao and Zhuhai, followed by Hong Kong and Macao, Guangzhou and Foshan (Fig. 9 a). Bilateral policy cooperation between Guangzhou

and Shenzhen was relatively lacking. The policy linkages between cities in the GBA were frequently interacted within the metropolitan area, and the policy linkages across metropolitan areas were relatively loose (Li et al., 2022). The multilateral policy links were mainly concentrated in Shenzhen metropolitan area and Guangzhou metropolitan area, which had the typical characteristics of national strategic drive and metropolitan strategic drive (Fig. 9 b). Multilateral policy links mainly focused

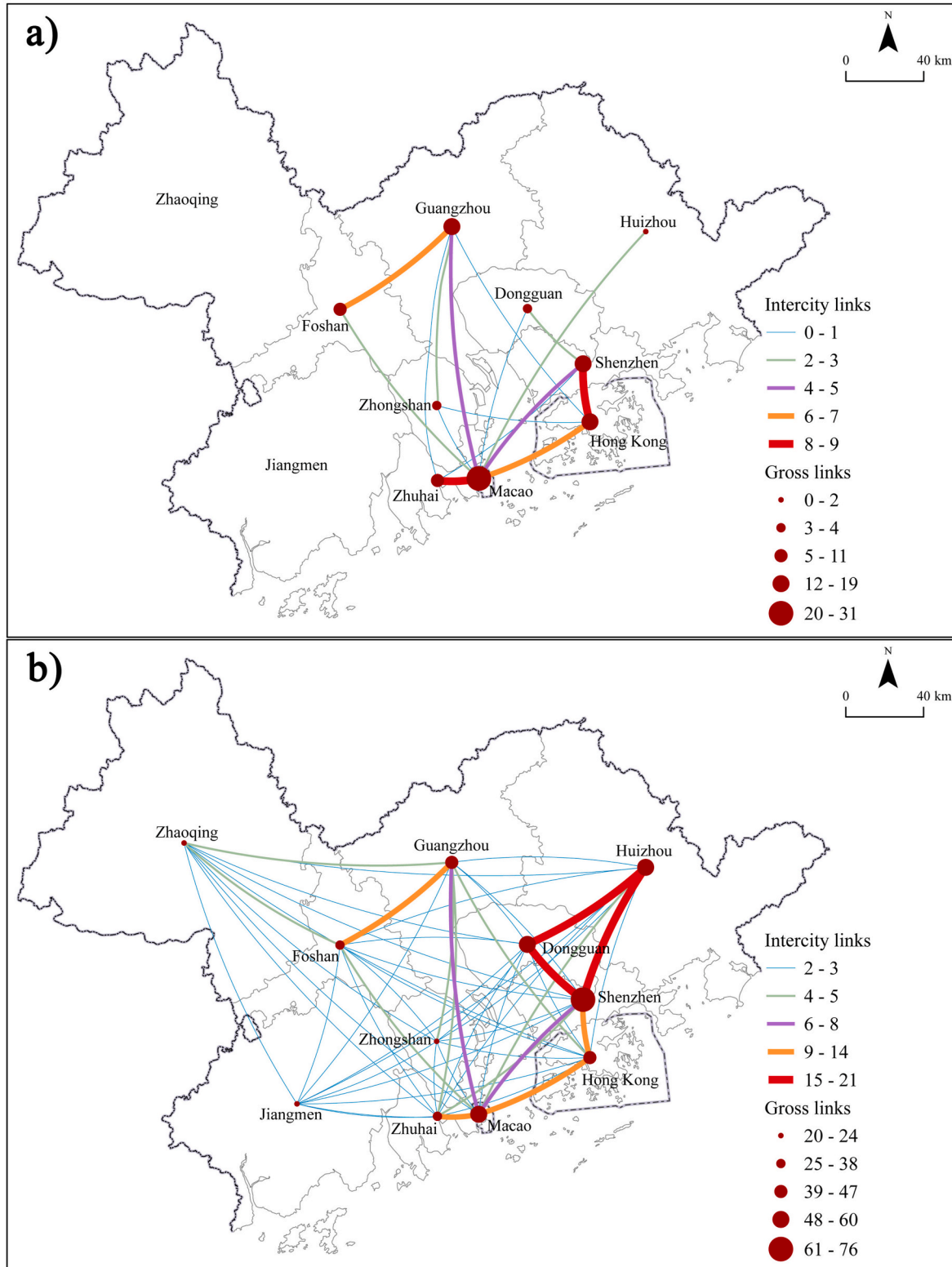


Fig. 9. Bilateral policy linkages (a) and multilateral policy linkages (b) in the GBA.

on infrastructure synergistic sharing policies, reflecting the common aspiration for a quality life circle that was suitable for living, working and traveling.

The joint meetings between cities were mainly reflected in Guangzhou and Shenzhen (Fig. 10 a), for example, the directors' joint meeting of the Standing Committee of the People's Congress of Guangzhou and Shenzhen was held in Guangzhou, the joint meeting of the tourism management departments and industry representatives of Guangzhou, Shenzhen and Zhuhai was held in Zhuhai, the Guangzhou Science and Technology Bureau and the Shenzhen Science and Technology Innovation Committee signed the Guangzhou-Shenzhen Science and Technology Innovation Cooperation Framework Agreement, and the Guangzhou Bureau of Industry and Information Technology and the Shenzhen Bureau of Industry and Information Technology reached a cooperation agreement on accelerating the joint cultivation of advanced automobile manufacturing cluster. The planning linkages between cities were mainly reflected in Guangzhou and Foshan, Guangzhou and Dongguan, Shenzhen and Dongguan, Shenzhen and Guangzhou, and more reflected in the planning linkages of transportation infrastructure between cities (Fig. 10 b). However, due

to the particularity of the GBA, the institutional arrangements of the participation subjects, interest demands, and implementation mechanisms were more complex and changeable. Therefore, it is urgent to establish an institutionalized wide-area administrative model for the cross-border regional synergistic development of the GBA (Liu et al., 2022).

3.5. Level of synergistic development in the Guangdong-Hong Kong-Macao greater bay area

The synergistic development between cities was measured from different dimensions, such as industrial synergy, innovation synergy, infrastructure synergy and institutional synergy, to achieve a hierarchical measurement of the synergistic development level of the GBA. Referring to the GII total score, namely the simple average of the input sub-index score and the output sub-index score, and through the simple average sum after standardization (the highest score of each dimension is 1 point, and the total is 4 points), the comprehensive measurement of the synergistic development level of the GBA was achieved (Fig. 11). The

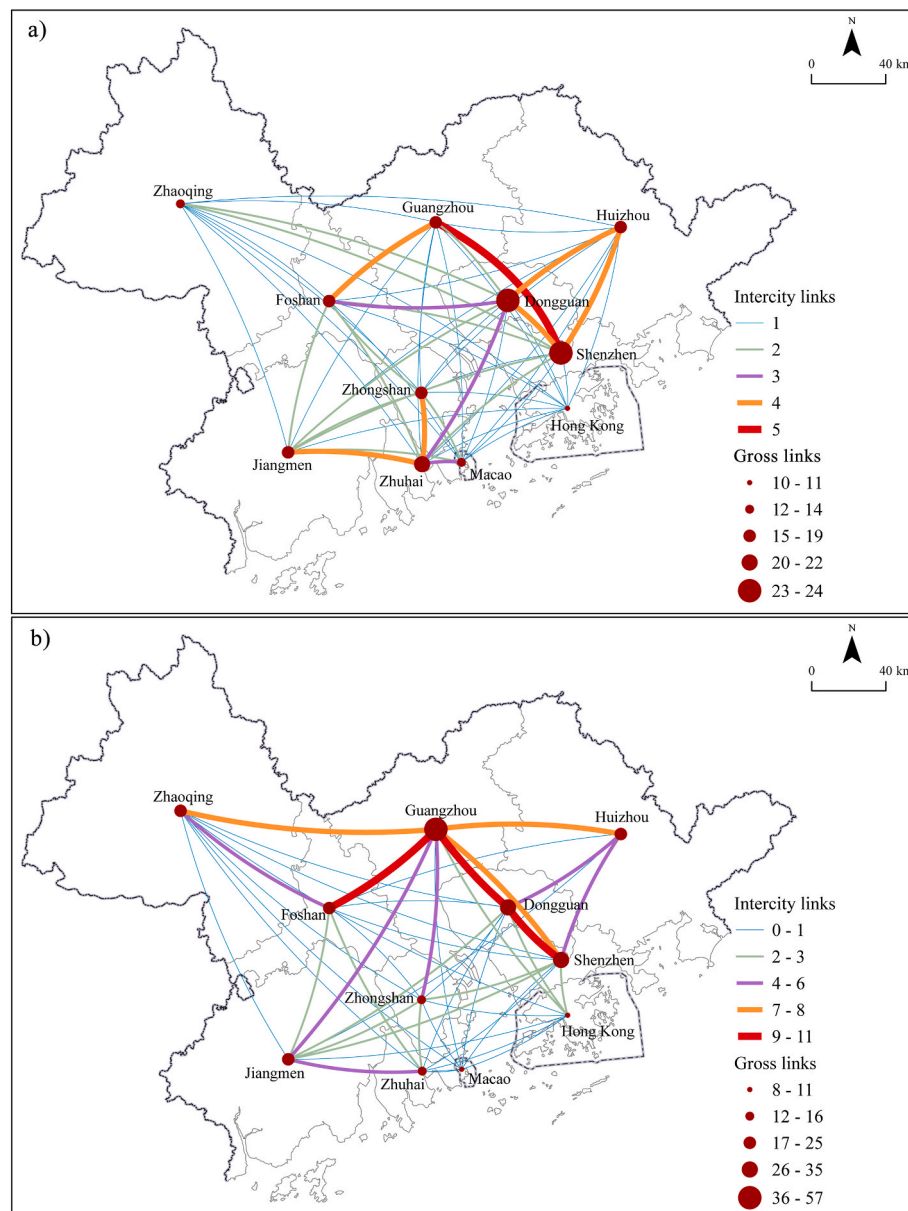


Fig. 10. Inter-city joint government meetings (a) and inter-city planning linkages (b) in the GBA.

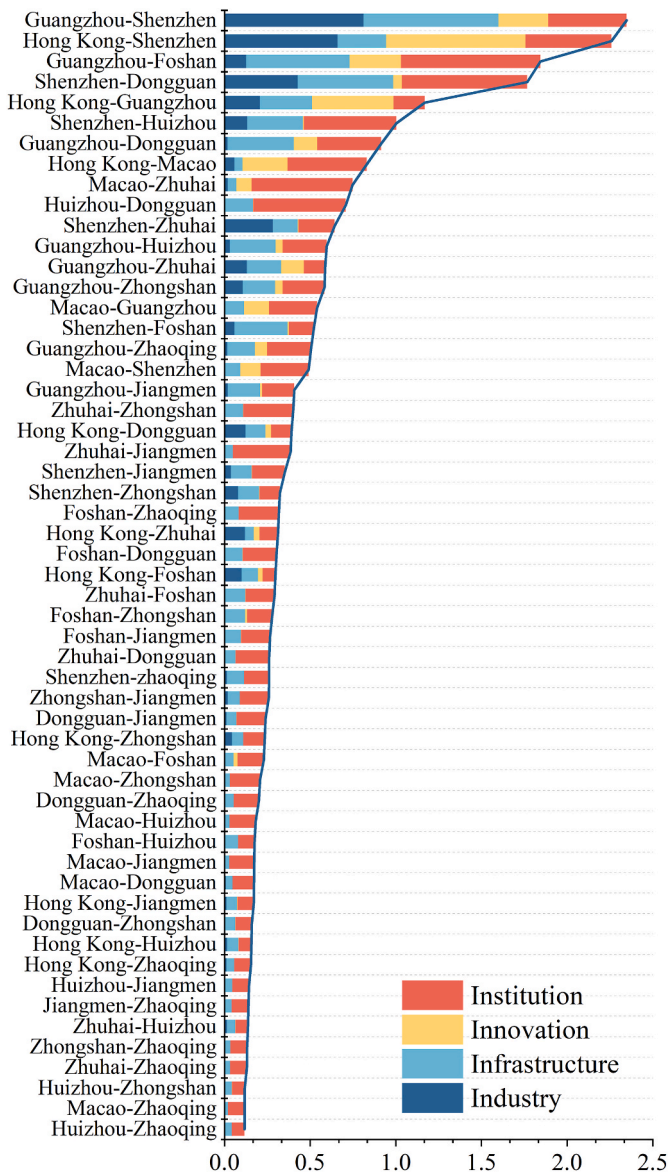


Fig. 11. Systematic evaluation of the synergistic development in the GBA.

relatively high level of synergistic development in the GBA is Guangzhou-Shenzhen, Hong Kong-Shenzhen, Guangzhou-Foshan, Shenzhen-Dongguan, Hong Kong-Guangzhou, Shenzhen-Huizhou, Guangzhou-Dongguan, Hong Kong-Macao, Macao-Zhuhai, Huizhou-Dongguan, and Shenzhen-Zhuhai. Guangzhou and Shenzhen have the highest level of synergistic development in the GBA due to their outstanding performance in the two dimensions of industrial synergy and infrastructure synergy. The synergistic development of Hong Kong and Shenzhen is more reflected in innovation synergy and industrial synergy. Guangzhou-Foshan and Shenzhen-Dongguan also have a high level of synergistic development due to their outstanding performance in the two dimensions of infrastructure synergy and institutional synergy, especially in the transportation links, information exchange, logistics links and financial links. Guangzhou-Dongguan and Shenzhen-Foshan also show a high level of synergistic development, which is outstanding in the dimension of infrastructure synergy, especially in the financial links and logistics linkage. At the same time, with the industrial penetration of Shenzhen and Foshan, compared with the solidified spatial structure of “Guangzhou-Foshan-Zhaozhou” and “Shenzhen-Dongguan-Huizhou”, the administrative barriers and boundary attributes between cities are gradually blurred. This remodeling of spatial

relationship will bring new opportunities and challenges to the synergistic development of the GBA. Compared with the synergistic development level of Hong Kong-Shenzhen, the synergistic development level of Macao-Zhuhai is relatively low. The synergistic development of Macao-Zhuhai is mainly reflected in institutional synergy, and industrial synergy and infrastructure synergy are biggest shortcomings of the current synergistic development of Macao and Zhuhai. Although the trend of de-bordering is obvious and the inter-city mobility is enhanced, the overall level of synergistic development of the GBA is not high. The level of synergistic development between Guangzhou and Shenzhen is 2.35, which is significantly lower than the overall score of 4 across the four different dimensions.

Drawing on the calculation method of GaWC's global city ranking by converting city links into total city connectivity, the connectivities between one city and 10 other cities in the GBA were summed up to obtain the synergistic development capacity of one city (Fig. 12).

The synergistic development capacity of Shenzhen and Guangzhou is the first echelon; Hong Kong, Dongguan and Foshan are the second echelon; Zhuhai, Macao and Huizhou are the third echelon; Zhongshan, Jiangmen and Zhaoqing are the fourth echelon. In general, the synergistic development capacity of cities in the GBA is mainly dominated by institutional synergy and infrastructure synergy, and the cities other than Guangzhou, Shenzhen and Hong Kong are insufficient in industrial synergy and innovation synergy. The two central cities of Guangzhou and Shenzhen have the strongest synergistic development capacity, and have advantages in all the four dimensions of industry, innovation, infrastructure and institution. Hong Kong and Macao, as two international cities, still have a lot of room for improvement in the synergistic development capacity within the GBA. Hong Kong has strong advantages in institutional synergy and innovation synergy, and Macao has strong advantages only in institutional synergy. Hong Kong and Macao should further transform their institutional advantages into competitive advantages in the process of implementing relevant policies, especially in industrial synergy and infrastructure synergy. Dongguan and Foshan have advantages in infrastructure synergy by virtue of the geographical proximity of Shenzhen-Dongguan metropolitan area and Guangzhou-Foshan metropolitan area, but should pay special attention to the deficiencies in industrial synergy and innovation synergy. Huizhou needs to further improve its ability in industrial synergy and innovation synergy. Zhuhai, Zhongshan and Jiangmen, as the key metropolitan areas on the west coast of the Pearl River, have obvious gaps in industrial synergy, innovation synergy and infrastructure synergy, and Zhongshan and Jiangmen also have obvious deficiencies in institutional synergy.

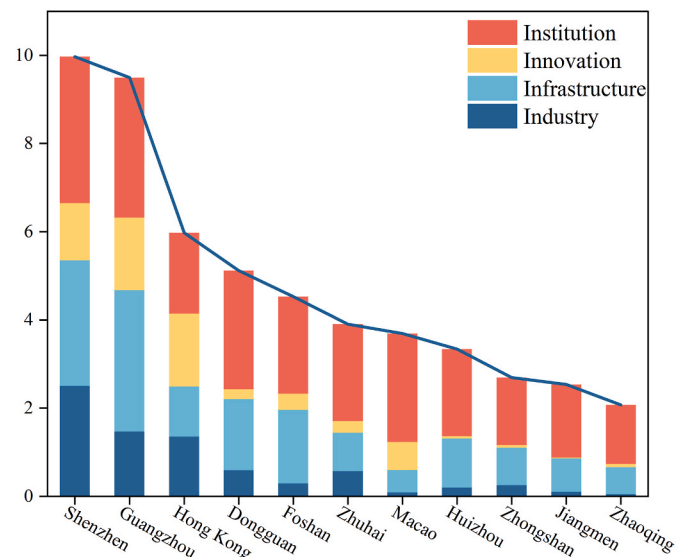


Fig. 12. The synergistic development capacity of different cities in the GBA.

Zhaoqing, as one of the key node cities in the GBA, has obvious gaps with other cities in the four dimensions of industry, innovation, infrastructure and institutions.

4. Conclusion and discussion

On the premise of fully understanding the scientific connotation and dynamic mechanism of the synergistic development of the urban agglomeration in the GBA, this paper constructed an evaluation index system from four dimensions of industrial synergy, innovation synergy, infrastructure synergy and institutional synergy. By comprehensively utilizing theories and methods such as “space of flow”, big data, network analysis and interlocking network model, this paper integrated multi-source heterogeneous massive data such as industrial enterprise data, venture capital data, intellectual property data, Amap data and government cooperation information data, focusing on measuring the one-to-one, one-to-many and many-to-many relationships between cities, and formed a comprehensive evaluation of the level and capacity of synergistic development. The main conclusions are as follows:

In terms of the dimensional evaluation of the synergistic development level of the GBA: Based on the perspective of industrial synergy, the capital flow between cities is mainly concentrated in Guangzhou-Shenzhen and Hong Kong-Shenzhen, and the industrial correlation between cities is mainly reflected in Hong Kong-Shenzhen, Guangzhou-Shenzhen. The knowledge flow between cities is concentrated in Hong Kong, Guangzhou and Shenzhen. From the perspective of transportation, communications, logistics, banking and other infrastructure, the infrastructure synergistic development level between Guangzhou and Shenzhen, Guangzhou and Foshan, and Shenzhen and Dongguan are higher. From the perspective of institutional collaboration arrangement, Hong Kong and Macao have institutional advantages in cross-border regional collaboration.

In terms of the comprehensive evaluation of the synergistic development level of the GBA: This area has gradually developed from the “core-periphery” spatial structure since the reform and opening up to an interconnected network structure. The areas with the best synergistic development level are Guangzhou and Shenzhen, Hong Kong and Shenzhen, Guangzhou and Foshan, Shenzhen and Dongguan. Meanwhile, Guangzhou and Dongguan, Shenzhen and Zhuhai, Shenzhen and Foshan show an interaction trend. Despite the enhanced inter-city mobility, the synergistic development level of the GBA is generally not high, and the average score of the synergistic development level is not higher than 2.4. In the context of Guangzhou metropolitan area and Shenzhen metropolitan area, Guangzhou and Dongguan, Shenzhen and Foshan show an interaction trend. These two super cities in close proximity, Guangzhou and Shenzhen, also show a new pattern of inter-city relationship.

Based on a novel perspective of the relationship network, this study proposes an indicator system and measurement method for collaboratively assessing the Guangdong-Hong Kong-Macao Greater Bay Area. It places greater emphasis on the mobility of knowledge, technology, capital, and other factors. This enables a scientific evaluation of the synergistic development level in the Guangdong-Hong Kong-Macao Greater Bay Area. The findings are significantly valuable for guiding assessments and decision-making regarding synergistic development situations in urban agglomerations and metropolitan areas within the region. Furthermore, they promote modernization in governance capacity within the area while enhancing synergistic governance levels in urban agglomerations. Due to data limitations, this study primarily relies on single-year data for comprehensive evaluations of synergistic development levels. Consequently, it fails to conduct dynamic analysis concerning regional synergistic development changes after implementing national strategies for constructing the Guangdong-Hong Kong-Macao Greater Bay Area. In future research endeavors, it is essential to continuously supplement and enhance both indicator systems and data accumulation that highlight distinctive characteristics specific to this

bay area region. This will enable dynamic assessments throughout construction processes from a collaborative perspective within the Guangdong-Hong Kong-Macao Greater Bay Area context. Additionally, further systematic research into synergistic mechanisms and paths is necessary with significant practical value towards establishing world-class bay areas and urban agglomerations.

CRedit authorship contribution statement

Changjian Wang: Conceptualization, Methodology, Formal analysis, Validation, Visualization, Writing – original draft. **Yuyao Ye:** Conceptualization, Data curation, Validation, Writing – review & editing. **Zhengdong Huang:** Data curation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Acuto, M., Parnell, S., Seto, K.C., 2018. Building a global urban science. *Nat. Sustain.* 1 (1), 2–4.
- Bodin, O., Chen, H., 2023. A network perspective of human–nature interactions in dynamic and fast-changing landscapes. *Natl. Sci. Rev.* 10 (7).
- Breul, M., 2020. Cities as regional nodes. In: *Global Value Chains: the Example of the Oil and Gas Industry in Southeast Asia, Gateway Cities in Global Production Networks: Insights from the Oil and Gas Industry in Southeast Asia*. Springer International Publishing, Cham, pp. 39–64.
- Burger, M.J., Meijers, E.J., 2016. Agglomerations and the rise of urban network externalities. *Pap. Reg. Sci.* 95 (1), 5–15.
- Castells, M., 2004. *The Network Society: A Cross-Cultural Perspective, Informationalism, Networks, and the Network Society: a Theoretical Blueprint*. Edward Elgar Publishing, pp. 1–3.
- Castells, M., 2009. The new economy: informationalism, globalization, networking. In: *The Rise of the Network Society*. Wiley-Blackwell, pp. 77–162.
- Che, B., Jin, C., Lu, Y., Zhu, C., 2021. Tracking the evolution of cooperative relationship networks among countries located along the Belt and Road: an application of massive cooperation events data. *Eurasian Geogr. Econ.* 1–28.
- Chen, S., 2018. Research on the spatial characteristics of factor flow and its international integration path in Guangdong-Hong Kong-Macao bay area. *Journal of South China Normal University (Social Science Edition)* (2), 27–32.
- Chong, Z., Pan, S., 2020. Understanding the structure and determinants of city network through intra-firm service relationships: the case of Guangdong-Hong Kong-Macao Greater Bay Area. *Cities* 103, 102738.
- Derudder, B., Taylor, P., Ni, P., De Vos, A., Hoyler, M., Hanssens, H., Bassens, D., Huang, J., Witlox, F., Shen, W., Yang, X., 2010. Pathways of change: shifting connectivities in the world city network, 2000–2008. *Urban Stud.* 47 (9), 1861–2008.
- Fan, Y., Fang, C., Zhang, Q., 2019. Coupling coordinated development between social economy and ecological environment in Chinese provincial capital cities-assessment and policy implications. *J. Clean. Prod.* 229, 289–298.
- Fang, C., 2017. Theoretical foundation and patterns of coordinated development of the Beijing-Tianjin-Hebei Urban Agglomeration. *Prog. Geogr.* 36 (1), 15–24.
- Fang, C., Yu, D., 2017. Urban agglomeration: an evolving concept of an emerging phenomenon. *Landsc. Urban Plann.* 162, 126–136.
- Fang, C., Yu, X., Zhang, X., Fang, J., Liu, H., 2020. Big data analysis on the spatial networks of urban agglomeration. *Cities* 102, 102735.
- Faulconbridge, J.R., 2019. Business services and the financing of global production networks: the case of global law firms in Southeast Asia. *J. Econ. Geogr.* 19 (4), 897–919.
- Gong, H., Hassink, R., Foster, C., Hess, M., Garretsen, H., 2022. Globalisation in reverse? Reconfiguring the geographies of value chains and production networks. *Camb. J. Reg. Econ. Soc.* 15 (2), 165–181.
- Grimm, N.B., Faeth, S.H., Golubiewski, N.E., Redman, C.L., Wu, J., Bai, X., Briggs, J.M., 2008. Global change and the ecology of cities. *Science* 319 (5864), 756–760.

- Hu, Z., Gong, J., Li, J., Li, R., Zhang, Z., Zhong, F., Wen, C., 2023. Valuing the coordinated development of urbanization and ecosystem service value in border counties. *J. Clean. Prod.* 415, 137799.
- Huh, H.-S., Park, C.-Y., 2021. A new index of globalisation: measuring impacts of integration on economic growth and income inequality. *World Econ.* 44 (2), 409–443.
- Ke, W., Xiao, B., Xu, K., et al., 2021. Fujian city networks spatial organization patterns based on railway passenger flow. *J. Fujian Normal Univ. (Philos. Soc. Sci. Ed.)* 37 (4), 87–98.
- Krugman, P., 1991. Increasing returns and economic geography. *J. Polit. Econ.* 99 (3), 483–499.
- Li, Q., Guo, L., Lei, Y., Cheng, C.i., Sima, X., 2022. Spatial connection analysis and coordinated development policy mechanism optimization in the Guangdong-Hong Kong-Macao greater bay area from the perspective of policy. *Trop. Geogr.* 42 (2), 269–282.
- Li, W., Yi, P., 2020. Assessment of city sustainability—coupling coordinated development among economy, society and environment. *J. Clean. Prod.* 256, 120453.
- Liu, J., 2017. Integration across a metacoupled world. *Ecol. Soc.* 22 (4).
- Liu, J., 2023. Leveraging the metacoupling framework for sustainability science and global sustainable development. *Natl. Sci. Rev.* 10 (7).
- Liu, J., Hull, V., Godfray, H.C.J., Tilman, D., Gleick, P., Hoff, H., Pahl-Wostl, C., Xu, Z., Chung, M.G., Sun, J., Li, S., 2018. Nexus approaches to global sustainable development. *Nat. Sustain.* 1 (9), 466–476.
- Liu, J., Mooney, H., Hull, V., Davis, S.J., Gaskell, J., Hertel, T., Lubchenco, J., Seto, K.C., Gleick, P., Kremen, C., Li, S., 2015. Systems integration for global sustainability. *Science* 347 (6225).
- Liu, W., Zou, J., 2014. A Direction of Regional Studies: Regional Interdependence, 01. *Areal Research and Development*, pp. 1–5. +16.
- Liu, X., Derudder, B., 2013. Analyzing urban networks through the lens of corporate networks: a critical review. *Cities* 31, 430–437.
- Liu, Y., Wang, Y., Li, H., 2020a. Industrial development of world-class bay areas and its enlightenment to the Guangdong-Hong Kong-Macao greater bay area. *Bull. Chin. Acad. Sci.* 35 (3), 312–321.
- Liu, Y., Yang, Y., Kang, L., Wang, Y., 2020b. Human-environment system in the Guangdong-Hong Kong-Macao greater bay area: global model and local response. *Geogr. Res.* 39 (9), 1949–1957.
- Liu, Y., Zhang, J., Wang, F., 2022. Scalar traps in the coordinated development of the Guangdong-Hong Kong-Macao greater bay area. *Prog. Geogr.* 41 (9), 1677–1687.
- Liu, Y., Zhang, X., Pan, X., Ma, X., Tang, M., 2020. The spatial integration and coordinated industrial development of urban agglomerations in the Yangtze River Economic Belt, China. *Cities* 104, 102801.
- Lu, D., 2017. Development of Pearl River Delta metropolitan Group and pan-Pearl River Delta economic cooperation zone. *Econ. Geography* 37 (4), 1–4.
- Ma, H., Li, Y., Huang, X., 2021. Proximity and the evolving knowledge polycentricity of megalopolitan science: evidence from China's Guangdong-Hong Kong-Macao Greater Bay Area, 1990–2016. *Urban Stud.* 58 (12), 2405–2423.
- Ma, H., Wei, Y.D., Dai, L., Xu, X., 2022. The proximity and dynamics of intercity technology transfers in the Guangdong-Hong Kong-Macao Greater Bay Area: evidence from patent transfer networks. *Environ. Plann.: Econ. Space* 54 (7), 1432–1449.
- Masuda, H., Kawakubo, S., Okitashi, M., Morita, K., 2022. Exploring the role of local governments as intermediaries to facilitate partnerships for the Sustainable Development Goals. *Sustain. Cities Soc.* 82, 103883.
- Moses, L.N., 1952. Regional Input-Output: A Method of Analysing Regional Interdependence.
- Peng, F., 2017. Economic spatial connection and spatial structure of Guangdong-Hong Kong-Macao greater bay and the surrounding area cities-an empirical analysis based on improved gravity model and social network analysis. *Econ. Geogr.* 37 (12), 57–64.
- Perroux, F., 1950. Economic space: theory and Applications I. *Q. J. Econ.* 64 (1), 89–104.
- Pichler, A., Diem, C., Brintrup, A., Lafond, F., Magerman, G., Buiten, G., Choi, T.Y., Carvalho, V.M., Farmer, J.D., Thurner, S., 2023. Building an alliance to map global supply networks. *Science* 382 (6668), 270–272.
- Qiu, J., Liu, Y., Chen, H., Gao, F., 2019. Urban network structure of Guangdong-Hong Kong-Macao greater bay area with the view of space of flows: A comparison between information flow and transportation flow. *Econ. Geogr.* 39 (6), 7–15.
- Rugman, A.M., Verbeke, A., 2004. A perspective on regional and global strategies of multinational enterprises. *J. Int. Bus. Stud.* 35 (1), 3–18.
- Shi, P., Song, C., Cheng, C., 2019. Geographical synergetics: from understanding human-environment relationship to designing human-environment synergy. *Acta Geograph. Sin.* 74 (1), 3–15.
- Taylor, P.J., 2001. Specification of the world city network. *Geogr. Anal.* 33 (2), 181–194.
- Taylor, P.J., 2005. Leading world cities: empirical evaluations of urban nodes in multiple networks. *Urban Stud.* 42 (9), 1593–1608.
- Taylor, P.J., Catalano, G., Walker, D.R.F., 2002. Measurement of the world city network. *Urban Stud.* 39 (13), 2367–2376.
- Taylor, P.J., Hoyler, M., Verbruggen, R., 2010. External urban relational process: introducing central flow theory to complement central place theory. *Urban Stud.* 47 (13), 2803–2818.
- Wang, C., Lu, M., Chen, J., Luo, H., Wang, F., 2022a. Construction and reconstruction of global value chain based on the perspective of urban networks: A case study of Huawei mobile phone suppliers. *Prog. Geogr.* 41 (9), 1606–1621.
- Wang, C., Wang, F., Zhang, X., Wang, Y., Su, Y., Ye, Y., Wu, Q., Zhang, H.o., 2022b. Dynamic features and driving mechanism of coal consumption for Guangdong province in China. *J. Geogr. Sci.* 32 (3), 401–420.
- Wang, C., Ye, Y., Wang, F., Huang, Z., Li, Q., Chen, Y., Lin, H., Wu, K., Lin, X., Zhang, H. o., 2022c. Measurement and evaluation of the coordinated development in the Guangdong-Hong Kong-Macao greater bay area. *Trop. Geogr.* 42 (2), 206–219.
- Wang, F., Luo, H., Wang, C., Ye, Y., Zhang, H.o., Lin, X., Chen, J., 2023. Spatial structure and influencing factors of urban network in the Guangdong-Hong Kong-Macao greater bay area from the financial relationships perspective. *Trop. Geogr.* 43 (4), 581–595.
- Webb, R., Bai, X., Smith, M.S., Costanza, R., Griggs, D., Moglia, M., Neuman, M., Newman, P., Newton, P., Norman, B., Ryan, C., Schandl, H., Steffen, W., Tapper, N., Thomson, G., 2018. Sustainable urban systems: Co-design and framing for transformation. *Ambio* 47 (1), 57–77.
- Wójcik, D., Keenan, L., Pažitka, V., Urban, M., Wu, W., 2022. The changing landscape of international financial centers in the twenty-first century: cross-border mergers and acquisitions in the global financial network. *Econ. Geogr.* 98 (2), 97–118.
- Xu, D., Ma, L., 2020. Restrictive factors and promoting approaches on collaborative ecological environment governance in the Guangdong-Hong Kong-Macao Greater Bay Area. *Geogr. Res.* 39 (9), 2165–2175.
- Xu, J., Du, Z., Yeh, A.G.O., 2023. Localization, regionalization, and domesticization of satellite industrial platform and urban transformation: a case study of Dongguan in the Pearl River Delta, China. *Cities* 139, 104368.
- Xu, J., Ye, Y., Luo, Z., Zhang, H.o., Wang, C., Wu, K., 2022. Multi-scalar spatial linkages initiated by high-tech industries in the Guangdong-Hong Kong-Macao Greater Bay Area and policy implications against the backdrop of China's new development paradigm. *Prog. Geogr.* 41 (9), 1592–1605.
- Yamasaki, K., Yamada, T., 2022. A framework to assess the local implementation of Sustainable Development Goal 11. *Sustain. Cities Soc.* 84, 104002.
- Yang, J., Jin, M., Chen, Y., 2023. Has the synergistic development of urban cluster improved carbon productivity? –Empirical evidence from China. *J. Clean. Prod.* 414, 137535.
- Ye, Y., Wang, X., Xu, J., Zhang, H.o., Liu, Z., Sun, J., Lu, Q., 2022. The connotation and mechanism changes of coordinated development of the Guangdong-Hong Kong-Macao greater bay area in the new period. *Trop. Geogr.* 42 (2), 161–170.
- Ye, Y., Wu, K., Xie, Y., Huang, G., Wang, C., Chen, J., 2019. How firm heterogeneity affects foreign direct investment location choice: micro-evidence from new foreign manufacturing firms in the Pearl River Delta. *Appl. Geogr.* 106, 11–21.
- Yeung, H.W.-c., 2021. Regional worlds: from related variety in regional diversification to strategic coupling in global production networks. *Reg. Stud.* 55 (6), 989–1010.
- Yeung, H.W.-c., 2022. Explaining geographic shifts of chip making toward East Asia and market dynamics in semiconductor global production networks. *Econ. Geogr.* 98 (3), 272–298.
- Yeung, H.W.-c., Coe, N.M., 2015. Toward a dynamic theory of global production networks. *Econ. Geogr.* 91 (1), 29–58.
- Yin, H., Zhang, Z., Wan, Y., Gao, Z., Guo, Y., Xiao, R., 2023. Sustainable network analysis and coordinated development simulation of urban agglomerations from multiple perspectives. *J. Clean. Prod.* 413, 137378.
- Yuan, B., Ren, S., Chen, X., 2017. Can environmental regulation promote the coordinated development of economy and environment in China's manufacturing industry? –A panel data analysis of 28 sub-sectors. *J. Clean. Prod.* 149, 11–24.
- Zhang, H., Geng, C., Wei, J., 2022. Coordinated development between green finance and environmental performance in China: the spatial-temporal difference and driving factors. *J. Clean. Prod.* 346, 131150.
- Zhang, L., Zhang, Q., Gao, Y., et al., 2022. Overall planning in dynamics: the spatiotemporal evolution characteristics of urban “Built-up Area-Population” coordinated development: analysis of chongqing since the time of direct administration. *J. Fujian Normal Univ. (Philos. Soc. Sci. Ed.)* 38 (3), 69–81.
- Zhang, W., Fang, C., Zhou, L., Zhu, J., 2020. Measuring megaregional structure in the Pearl River Delta by mobile phone signaling data: a complex network approach. *Cities* 104, 102809.
- Zheng, Y., He, Y., Zhou, Q., Wang, H., 2022. Quantitative evaluation of urban expansion using NPP-viirs nighttime light and landsat spectral data. *Sustain. Cities Soc.* 76, 103338.
- Zhou, C., Deng, H., Shi, C., 2018. A study on synergic development of Guangdong-Hong Kong-Macao greater bay area. *Planner* 34 (4), 5–12.